Lecture II - Explore Vs. Exploit

Programming: Everyday Decision-Making Algorithms

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Explore Vs. Exploit

You have 45 minutes for lunch. Do you try the new restaurant or go to your reliable favorite?

. . .

📦 Quick Poll: Raise your hand, who would try the new restaurant?

. . .

This decision happens hundreds of times daily — and there's optimal math behind it!

Photo by Jillian Amatt on Unsplash

Core Concepts

🔍 Explore

- Definition: Gathering new information
- Risk: Unknown outcomes
- Example: Trying that new sushi place

🎯 Exploit

- Definition: Using known information
- Risk: Missing better options
- Example: Your go-to pizza joint

> ❗⚖️ The Fundamental Trade-off
>
> These are mutually exclusive choices in any given moment. You can't have your cake and eat it (too).

Question: Do you have any examples of explore vs exploit in your life?

🏥 Medical: Clinical Trials

Explore: Test experimental drug on new patients

- Risk: Unknown side effects, potential harm
- Reward: Breakthrough treatment discovery

Exploit: Use proven treatment

- Risk: Missing better cures
- Reward: Predictable, safe outcomes

💻 Tech: A/B Testing

Explore: Test new website design

- Cost: Development time, potential lost conversions
- Benefit: Higher conversion rates, better UX

Exploit: Keep current design

- Cost: Missed optimization opportunities
- Benefit: Stable, known performance

💕 Personal: Dating Apps

Explore: Meet someone new

- Risk: Bad dates, wasted time
- Reward: Finding “the one”

Exploit: Meet someone you already know

- Risk: Settling for “good enough”
- Reward: Comfortable, known connection

The Daily Decision Dilemma

Explore Moments

- Try a new coffee shop
- Learn a new skill
- Watch a different genre
- Take a new route to work

Exploit Moments

- Order your usual drink
- Use familiar tools
- Rewatch favorite shows
- Take the fastest route

> 💡 The Million Dollar Question
>
> What is the optimal balance? Science has been asking this for over 50 years!

The problem with exploitation

Question: Any ideas what it might be?

- Exploitation is not always the best strategy.
- Especially when you have very limited information.

- When you stop exploring, you might miss better options.
- Imagine you are not able to gather new information and could only choose known alternatives.

The problem with exploration

Question: Any ideas what it might be?

- Exploration is not always the best strategy.
- Especially when you are limited in using new information.
- When you constantly explore, you might never enjoy the fruits of your exploration.
- Imagine you could only eat each meal only once, hear each song only once, talk to each person only once…

Multi-Armed Bandit Problem

Decision Support

- To provide support, computer scientists formulated the explore-exploit trade-off.
- It is known as the multi-armed bandit problem.
- Question: What is a one-armed bandit?

One Armed Bandit

Photo by Kabir M on Unsplash

Multi-Armed Bandit Problem

Imagine: You walk into a casino with 3 slot machines

- Each machine has a different (unknown) win rate
- You have $100 and 20 pulls total
- Goal: Maximize your winnings

How is this related to explore vs. exploit?

- Explore: Try different machines
- Exploit: Play the best-performing machine

Multi-Armed Bandit Problem II

- By playing a slot machine, the gambler can gather information about the probability of winning.
- But each pull of a lever comes with a certain cost.
- It's the aim of the gambler to maximize his winnings.

The Core Question

How do you decide which machine to play next?

Interactive Casino Challenge

Your Current Situation

You've been playing for a while. Here's what you know:

Machine	Pulls	Wins	Win Rate	Your Choice?
A 🎰	15	9	60%	
B 🎰	2	1	50%	
C 🎰	0	0	???	

Class Vote: Raise your hand for your choice!

- Machine A (known performer)
- Machine B (limited data)
- Machine C (complete unknown)

Casino Challenge Follow-up

Your Current Situation

You've been playing for a while. Here's what you know:

Machine	Pulls	Wins	Win Rate	Your Choice?
A 🎰	15	9	60%	
B 🎰	2	1	50%	
C 🎰	0	0	???	

ℹ Follow-up Questions

- What if you only had 2 pulls left?
- What if you had 100 pulls left?
- How does this change your strategy?

Expected value for a decision

- The expected value of a slot machine is the average number of wins per pull.
- Expected value of machine A = E(A) = 9/15 = 0.6
- Expected value of machine B = E(B) = 1/2 = 0.5
- Machine A has the higher expected value.
- But 2 and even 15 pulls are not a large number (considering standard deviation).

Multi-Armed Bandit III

- The multi-armed bandit problem represents a lot of different real-world decisions.
- It shows us, that there might be a difference between the optimal long-term average performance and the optimal immediate performance.
- Which lever to pull next depends completely on something we haven't discussed yet:

Multi-Armed Bandit IV

- How long you plan to be in the casino?
- Question: Why is this important?
- Question: How does this influence our decision on taking machine A or machine B?

. . .

> "I'm more likely to try a new restaurant when I move to a city than when I'm leaving it" (Chris Stucchio)

The influence of the interval

- Let's call the time you plan to be in the casino "the interval".
- The longer the interval, the more (in general) you should explore, since you will have more opportunities to exploit the gathered information.
- The shorter the interval, the more you should exploit your current knowledge.
- The optimal strategy depends on the length of the interval.

Interval and Exploration

Explore when you have the time to use the resulting knowledge.

> "I moved to Pune, India, and I just [...] eat everywhere that didn't look like it was gonna kill me" (Chris Stucchio)

Exploration

Photo by Colin Maynard on Unsplash

Interval and Exploitation

Exploit when you are ready to cash in.

> "And as I was leaving the city I went back to all my old favorites, rather than trying out new stuff [...]. Even if I find a slightly better place, I'm only going to go there once or twice, so why take the risk?" (Chris Stucchio)

Exploitation

Photo by Cristina Gottardi on Unsplash

Hollywood's Strategy

ℹ Sequels in Top 10 Highest-Grossing Movies

Year	Number of Sequels
1981	2
1991	3
2001	5
2011	8

💡 💭 Class Discussion

What do you think:

- What explains this trend?
- How does this relate to explore vs. exploit?
- What does this suggest about Hollywood's "time horizon"?

Sequels…

. . .

The Insight: Studios are exploiting proven franchises because they think they're approaching the "end of their interval" (streaming disruption, changing audience preferences).

Photo by Universal Pictures on Wikipedia

A Possible Explanation

- Making a brand new movie is risky but has the potential to create a new fan base. (explore)
- From a Studio's perspective, a sequel is a movie with a guaranteed fan base, a cash cow, a sure thing, an exploit.
- One possible explanation for the numbers is that the studios think they are approaching the end of their interval.
- They are pulling the arms of the best machines they've got before the casino turns them out.

Multi-Armed Bandit VI

- While the so far provided anecdotes are helping us to understand the explore-exploit trade-off, they are far away from beeing a satisfying "optimal" solution.
- Actually, finding an algorithm that tells us exactly how to to handle the trade-off is a very hard problem.
- On the way there were many interesting approaches…

Win-Stay Lose-Shift

Win-Stay Lose-Shift[1]

- Question: What do you think, what the win-stay lose-shift strategy does?
- The win-stay lose-shift strategy is a simple strategy that is often used in multi-armed bandit problems.
- Based on the idea that if you have won, you should stay. If you have lost, you should switch.
- This strategy is not always the best strategy, but it is simple and proven better than choose an arm at random.

Win-Stay = No Brainer?

- Question: What do you think about win-stay?
- If you decide to pull an arm and you win, you should pull the same arm again.
- Nothing changes, except the attractiveness of the arm you just pulled is higher.

Lose-Shift is another story

- Question: What do you think about lose-shift?
- Changing arms each time you lose might be a prety rush move.
- Imagine you're eating at your favorite restaurant for the tenth time in a row.
- You have always been very satisfied (win), but today you are disappointed (loose).
- Should you turn your back on the restaurant?

Like most of the time, easy answers comes with problems

- The win-stay lose-shift strategy penalizes losses too much.
- The strategy does not take into account the interval.
- But the strategy was good start to develop better approaches.

The Bellman Approach

The Bellman approach I

- Few years later, Richard Bellman, found an exact solution for all cases with finite and known intervals.
- Bellman found that under assumptions, exploit vs explore can be seen as an optimal stopping problem.
- The question is, when to stop exploring and start exploiting.
- The solution is based on dynamic programming and backward calculation starting from the final pull (analogous to the secretary problem with full information).

The Bellman approach II

Bellman found that the following equation provides the optimal strategy (when the assumptions hold):

[1]For more details see Robbins, H. (1952) 'Some aspects of the sequential design of experiments', Bulletin of the American Mathematical Society, 58.

$$\mathbb{E}[\cdot] = \frac{w+1}{w+l+2}$$

where w is the number of wins and l is the number of loses.

Question time

$$\mathbb{E}[\cdot] = \frac{w+1}{w+l+2}$$

- Question: What is E[B]?
- Question: What is E[A]?
- Question: What machine shall we play according to the Bellman approach?

The Bellman approach III

The following table shows the expected value for different win-lose scenarios.

Loses/Wins	1	2	3	4	5	6	7	8	9	10
1	0.50	0.60	0.67	0.71	0.75	0.78	0.80	0.82	0.83	0.85
2	0.40	0.50	0.57	0.63	0.67	0.70	0.73	0.75	0.77	0.79
3	0.33	0.43	0.50	0.56	0.60	0.64	0.67	0.69	0.71	0.73
4	0.29	0.38	0.44	0.50	0.55	0.58	0.62	0.64	0.67	0.69
5	0.25	0.33	0.40	0.45	0.50	0.54	0.57	0.60	0.63	0.65
6	0.22	0.30	0.36	0.42	0.46	0.50	0.53	0.56	0.59	0.61
7	0.20	0.27	0.33	0.38	0.43	0.47	0.50	0.53	0.56	0.58
8	0.18	0.25	0.31	0.36	0.40	0.44	0.47	0.50	0.53	0.55
9	0.17	0.23	0.29	0.33	0.38	0.41	0.44	0.47	0.50	0.52
10	0.15	0.21	0.27	0.31	0.35	0.39	0.42	0.45	0.48	0.50

The Bellman approach IV

- The calculation of the optimal strategy is very extensive when there are many arms and a long interval.
- And yet the approach does not help us in many scenarios because we do not know the exact length of the interval (time in the casino).
- At this point, it looked like the multi-armed bandit problem would remain a problem without a solution.

[2] For more details see Gittins, J. (1979) 'Bandit Processes and Dynamic Allocation Indices', Journal of the Royal Statistical Society. Series B (Methodological).

The Gittins Index

The Gittins Index I[2]

- In the 1970s Unilever asked a young mathematician, John Gittins, to help optimize their drug trials.
- Given different compounds, what is the quickest way to determine which is likely to be effective?
- Gittins found an optimal strategy and abstracted the problem to a general level.
- He found the solution to the multi-armed bandit problem.

The Gittins Index II

- A major problem with the multi-armed banded problem is that previous solutions made very critical assumptions about the underlying interval.
- For example, that the length of the interval is known at the beginning of the analysis.
- In his approach, future wins (e.g., cash flows) are discounted so that any interval length (including infinity) can be considered.
- Geometric discounting assumption, but the approach can be extended to other discounting assumptions.

What is Discounting?

- Future rewards are valued less than immediate rewards.
- Future rewards are weighted by a discount factor.
- Monetary context: Easy to justify discounting:
  - Interest rates: Money today can earn interest.
  - Opportunity costs: Immediate use of money has value.
- Non-monetary context: Discounting also makes sense:
  - Time: Immediate gratification often preferred.
  - Example: Tonight's dinner vs. the same dinner in a week's time.

The Gittins Index III

- The Gittins index can be used for any problems of the form of the multi-armed bandit problem.
- That means it solves the explore-exploit trade-off.
- Let's consider our machine A and B example one last time.
  - Machine A: 15 pulls, 9 wins, 6 loses.
  - Machine B: 2 pulls, 1 win, 1 lose.

The Gittins Index IV

Loses/Wins	0	1	2	3	4	5	6	7	8	9
0	.7029	.8001	.8452	.8723	.8905	.9039	.9141	.9221	.9287	.9342
1	.5001	.6346	.7072	.7539	.7869	.8115	.8307	.8461	.8588	.8695
2	.3796	.5163	.6010	.6579	.6996	.7318	.7573	.7782	.7956	.8103
3	.3021	.4342	.5184	.5809	.6276	.6642	.6940	.7187	.7396	.7573

Loses/Wins	0	1	2	3	4	5	6	7	8	9
4	.2488	.3720	.4561	.5179	.5676	.6071	.6395	.6666	.6899	.7101
5	.2103	.3245	.4058	.4677	.5168	.5581	.5923	.6212	.6461	.6677
6	.1815	.2871	.3647	.4257	.4748	.5156	.5510	.5811	.6071	.6300
7	.1591	.2569	.3308	.3900	.4387	.4795	.5144	.5454	.5723	.5960
8	.1413	.2323	.3025	.3595	.4073	.4479	.4828	.5134	.5409	.5652
9	.1269	.2116	.2784	.3332	.3799	.4200	.4548	.4853	.5125	.5373

Question

Choose machine A or B according to the Gittins index?

- The index for machine B (0.6346) is higher than for machine A (0.6300).
- The index shows a clear win-stay pattern.
- There is a relaxed lose-shift pattern.
- At the (0,0) point we see the exploration bonus (premium).
- The index converges to 1/2 for a 50/50 chance game.

Gittins Index: The Scary Math

😱 The Complex Formula

$$G_i(s_i, f_i) := \sup_{\tau \geq 1} \frac{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t \cdot r_i^t \,\middle|\, s_i, f_i\right]}{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t\right]}$$

- Extremely complex to calculate by hand
- Requires advanced mathematics (optimization theory)
- Calculation not practical for everyday decisions

> ! Important
>
> You don't need to understand the math - just use the lookup table!

What It Actually Means

- Core idea: "What's the best possible average reward from this option?"
- Higher Index = Better Choice
- Example: Machine A (9 wins, 6 losses): Index = 0.6300; Machine B (1 win, 1 loss): Index = 0.6346
- Machine B wins due to exploration bonus for less-tested options
- The Gittins Index automatically balances exploration and exploitation

Explore vs Exploit: Summary

Explore vs Exploit: Summary

- Consider an explore vs exploit decision situation.
- As you learned exploiting comes with a known (expected) outcome for example E(exploit) = X
- Exploring comes with an unknown outcome E(explore) = ?
- What should you do according to decision science?

Explore vs Exploit: Anecdotally

- If you have a long interval, you should explore, choose B untill you are sure about E(explore).
- If you have a short interval, you should exploit, choose A.

Explore vs Exploit: Mathematically

- If E(exploit) and E(explore) are known, choose higher expected value.
- If E(explore) is unknown, but you know the length of the interval, the Bellman-approach provides the optimal strategy.
- If E(explore) is unknown, and you do not know the length of the interval, the Gittins index provides the optimal strategy.

🎯 Your Personal Action Plan

💡 Rate yourself (1-10) on each:

- Career exploration: Trying new skills/roles vs. deepening current expertise
- Social exploration: Meeting new people vs. investing in current relationships
- Learning exploration: New subjects vs. mastering current interests
- Life exploration: New experiences vs. enjoying familiar pleasures

! Key Insights to Remember

1. Exploration has inherent value - it increases your chance of finding the best
2. Time horizon matters - more time left = more exploration optimal
3. Age affects strategy - younger people should explore more
4. Don't stop exploring completely - even experts need some exploration

Takeaways

Your todays takeaway from the lecture should be: Be sensitive to how much time you have left in the casino and explore, explore, explore…

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[3]
- Ferguson, T.S. (1989) 'Who solved the secretary problem?', Statistical Science, 4(3). doi:10.1214/ss/1177012493.

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

...

> **Note**
>
> Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

More Literature

For more interesting literature, take a look at the literature list of this course.

[3] The main inspiration for this lecture. Nils and I have read it and discussed it in depth, always wanting to translate it into a course.